

UNIT – IV

MULTIPLE INTEGRALS

Complete Exam Guide with Theory, Formulas, Diagrams & Solved Examples

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Prepared for MSBTE K-Scheme Students | Exam Ready Reference

4.1 Introduction to Multiple Integrals

In Engineering Mathematics, we often deal with functions that depend on **more than one variable** — for example, area, volume, mass, and centre of gravity all require integration over two or three dimensions. Multiple integrals are a systematic way to perform such integrations.

When a physical quantity is spread over a region in 2D or 3D space, we cannot compute it with a single integral. We need to integrate over **two variables simultaneously** (Double Integral) or **three variables simultaneously** (Triple Integral).

Key Keywords

- ★ **Double Integral** — Integration over a 2D region R ; denoted $\iint_R f(x,y) \, dA$.
- ★ **Triple Integral** — Integration over a 3D volume V ; denoted $\iiint_V f(x,y,z) \, dV$.
- ★ **Region of Integration** — The 2D or 3D domain over which integration is performed.
- ★ **Iterated Integral** — A double/triple integral evaluated step-by-step, one variable at a time.
- ★ **Limits of Integration** — The boundary values of each variable — can be constants or functions.
- ★ **Cartesian Co-ordinates** — Standard x - y (or x - y - z) system for expressing regions.
- ★ **Polar Co-ordinates** — System using (r, θ) — useful for circular/symmetric regions.
- ★ **Change of Order** — Reversing the order of integration to simplify evaluation.

■ **Note:** Every double integral can be viewed as a limit of a double sum — just like a single integral is the limit of a Riemann sum.

4.2 Double Integration in Cartesian Co-ordinates

What is a Double Integral?

Let $f(x, y)$ be a continuous function defined over a closed region R . We divide R into n small sub-regions of areas $A_1, A_2, \dots, A_n$, pick any point (x_i, y_i) inside each sub-region, and form the sum $\sum f(x_i, y_i) A_i$. As $n \rightarrow \infty$ and each area $A_i \rightarrow 0$, this sum converges to the **Double Integral**:

$$\iint_R f(x, y) dA = \lim_{(n \rightarrow \infty)} \sum f(x_i, y_i) A_i$$

In practice, the double integral is written as an **iterated integral** and evaluated by performing two successive single integrations:

$$\iint f(x, y) dy dx \text{ or } \iint f(x, y) dx dy$$

How to Evaluate a Double Integral

Step 1: Start with the **inner integral**. Integrate $f(x, y)$ with respect to the inner variable (say y), treating the outer variable (x) as a **constant**.

Step 2: Substitute the **inner limits** to get a function of the outer variable only.

Step 3: Integrate the result with respect to the **outer variable** between the outer limits.

Step 4: Substitute the outer limits to get the **final numerical answer**.

Region of Integration – Two Cases

The shape of the region R determines how we set up the limits. There are **two standard cases**:

Case	Integral Form	Strip Type	y-limits	x-limits
(i)	$\int_a^b \int_{f_1(x)}^{f_2(x)} f dy dx$	Vertical strip	$y = f_1(x) \text{ to } y = f_2(x)$	$x = a \text{ to } x = b$
(ii)	$\int_c^d \int_{g_1(y)}^{g_2(y)} f dx dy$	Horizontal strip	$y = c \text{ to } y = d$	$x = g_1(y) \text{ to } x = g_2(y)$

Double Integration – Region of Integration (Cartesian)

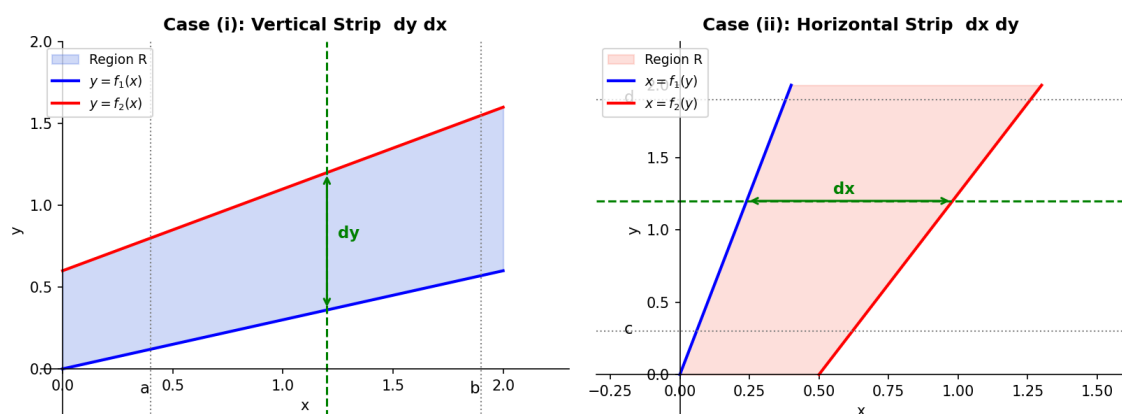


Figure 4.1 — Left: Vertical strips for $dy dx$ order. Right: Horizontal strips for $dx dy$ order.

Important Rule About Correct Form

■ **Important:** If the limits of the inner integral are functions of x , then the first integration must be with respect to y . If the limits of the inner integral are functions of y , the first integration must be with respect to x . Always check the correct form before evaluating!

Key Formulas

Standard Double Integral:

$$\int_a^b \int_{f_1(x)}^{f_2(x)} f(x,y) dy dx$$

When limits are constants:

$$\int_a^b \int_c^d f(x,y) dy dx = \left[\int_c^d f(x,y) dy \right] \text{ evaluated from } a \text{ to } b$$

Separable functions:

$$\int_a^b \int_c^d f(x) \cdot g(y) dy dx = \left[\int_a^b f(x) dx \right] \times \left[\int_c^d g(y) dy \right]$$

Solved Examples

→ **Example:** Evaluate $\int_0^1 \int_0^2 x(x+y) dy dx$

Solution:

Inner integration (w.r.t. y , treating x as constant):

$$\int_0^2 x(x+y) dy = [x^2y + xy^2/2]_0^2 = (2x^2 + 2x) - (x^2 + x/2) = x^2 + 3x/2$$

Outer integration (w.r.t. x):

$$\int_0^1 (x^2 + 3x/2) dx = [x^3/3 + 3x^2/4]_0^1 = 1/3 + 3/4 = 13/12$$

Answer: 13/12

→ **Example:** Evaluate $\int_0^1 \int_0^1 xy(x-y) dy dx$

Solution:

Inner integration (w.r.t. y):

$$\int_0^1 (x^2y - xy^2) dy = [x^2y^2/2 - xy^3/3]_0^1 = b^2x^2/2 - b^3x/3$$

Outer integration (w.r.t. x):

$$\int_0^1 (b^2x^2/2 - b^3x/3) dx = [b^2x^3/6 - b^3x^2/6]_0^1 = a^3b^2/6 - a^2b^3/6$$

Answer: $(a^3b^2/6)(a - b)$

→ **Example:** Evaluate $\int_0^3 \int_0^2 e^{x+y} dy dx$

Solution (Separable):

Since $e^{(x+y)} = e^x \cdot e^y$, the function is separable:

$$\begin{aligned}\int_1^3 \int_1^2 e^{x+y} dy dx &= \left[\int_1^3 e^x dx \right] \times \left[\int_1^2 e^y dy \right] \\ &= [e^x]_1^3 \times [e^y]_1^2 = (e^3 - 1)(e^2 - 1)\end{aligned}$$

Answer: $(e^3 - 1)(e^2 - 1)$

4.3 Double Integration in Polar Co-ordinates

Why Polar Co-ordinates?

When the region of integration is **circular, annular, or has angular symmetry**, expressing the integral in Cartesian form becomes very complicated. In such situations, **Polar Co-ordinates** make the limits much simpler and the integral easier to evaluate.

Standard Form of Polar Double Integral

$$\int_{\theta_1}^{\theta_2} \int_{r_1(\theta)}^{r_2(\theta)} f(r, \theta) r \, dr \, d\theta$$

The extra factor "**r**" in the integrand comes from the Jacobian of the polar transformation. It is **always required** when converting from Cartesian to Polar form.

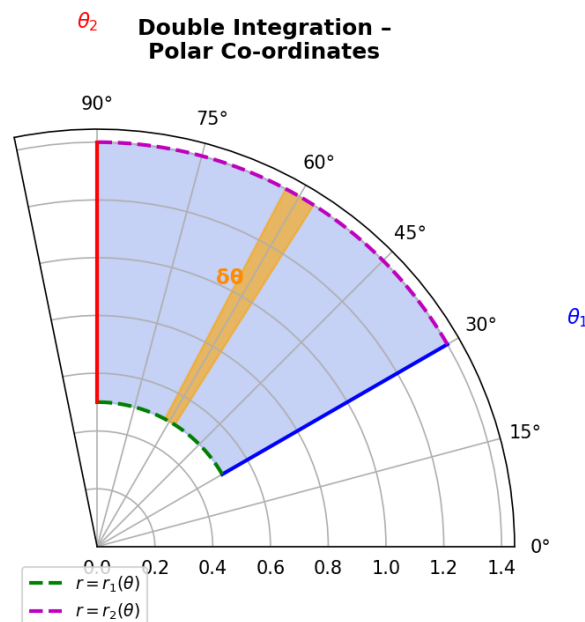


Figure 4.2 — Region bounded by $r_1(\theta) \leq r \leq r_2(\theta)$ and $\theta_1 \leq \theta \leq \theta_2$. PQ is a wedge of angular thickness $\delta\theta$.

How to Evaluate a Polar Double Integral

- Step 1:** Write limits: θ varies between two angles (θ_1 and θ_2), and r varies between two curves $r_1(\theta)$ and $r_2(\theta)$.
- Step 2:** Integrate $f(r, \theta) \cdot r$ with respect to r first (inner integral), treating θ as constant.
- Step 3:** Integrate the resulting function with respect to θ (outer integral).

Key Formulas Used in Polar Integration

Walli's Formula – $\sin^2\theta$ or $\cos^2\theta$:

$$\int_0^{\pi/2} \sin^2 \theta \, d\theta = \int_0^{\pi/2} \cos^2 \theta \, d\theta = (1/2) \cdot (\pi/2)$$

Walli's Formula – $\cos^3 \theta$ or $\sin^3 \theta$:

$$\int_0^{\pi/2} \cos^3 \theta \, d\theta = \int_0^{\pi/2} \sin^3 \theta \, d\theta = (2/3)$$

Walli's Formula – $\cos^4 \theta$:

$$\int_0^{\pi/2} \cos^4 \theta \, d\theta = (3/4) \cdot (1/2) \cdot (\pi/2)$$

Symmetry shortcut:

$$\int_{-\pi/2}^{\pi/2} f(\theta) \, d\theta = 2 \int_0^{\pi/2} f(\theta) \, d\theta \text{ (when } f \text{ is even in } \theta)$$

Solved Examples

⇒ **Example:** Evaluate $\int_0^{\pi/2} \int_0^{\sin \theta} r \, dr \, d\theta$

Solution:

$$\text{Inner: } \int_0^{\sin \theta} r \, dr = [r^2/2]_0^{\sin \theta} = \sin^2 \theta / 2$$

$$\text{Outer: } \int_0^{\pi/2} \sin^2 \theta / 2 \, d\theta = (1/2) \cdot (1/2) \cdot (\pi/2) = \pi/8$$

Answer: $\pi/8$

⇒ **Example:** Evaluate $\int_0^{\pi} \int_0^{\sin \theta} r \, dr \, d\theta$

Solution:

$$\text{Inner: } \int_0^{\sin \theta} r \, dr = \sin^2 \theta / 2$$

$$\text{Outer: } \int_0^{\pi} \sin^2 \theta / 2 \, d\theta = (1/2) \cdot \int_0^{\pi} (1 - \cos 2\theta) / 2 \, d\theta = \pi/4$$

Answer: $\pi/4$

4.4 Change of Order of Integration

Why Change the Order?

Sometimes the given integral is **difficult or impossible to evaluate** in the given order of integration. By changing the order (from $dy\,dx$ to $dx\,dy$, or vice versa), we often get a simpler integral that can be evaluated easily. This technique is called the **Change of Order of Integration**.

Step-by-Step Working Rule

- Step 1: Identify the given order** of integration ($dy\,dx$ or $dx\,dy$).
- Step 2: Write the equations** corresponding to the given limits (e.g., $y = x$, $y = a$, $x = 0$, $x = a$).
- Step 3: Sketch the region of integration** by plotting these boundary curves and lines.
- Step 4: Choose the new order** of integration. If the original was $dy\,dx$, the new order will be $dx\,dy$, and vice versa.
- Step 5: Draw the appropriate strip**: vertical strip for $dy\,dx$, horizontal strip for $dx\,dy$.
- Step 6: Determine the new limits** by reading off the strip from the region diagram.
- Step 7: Evaluate the integral** with the new limits.

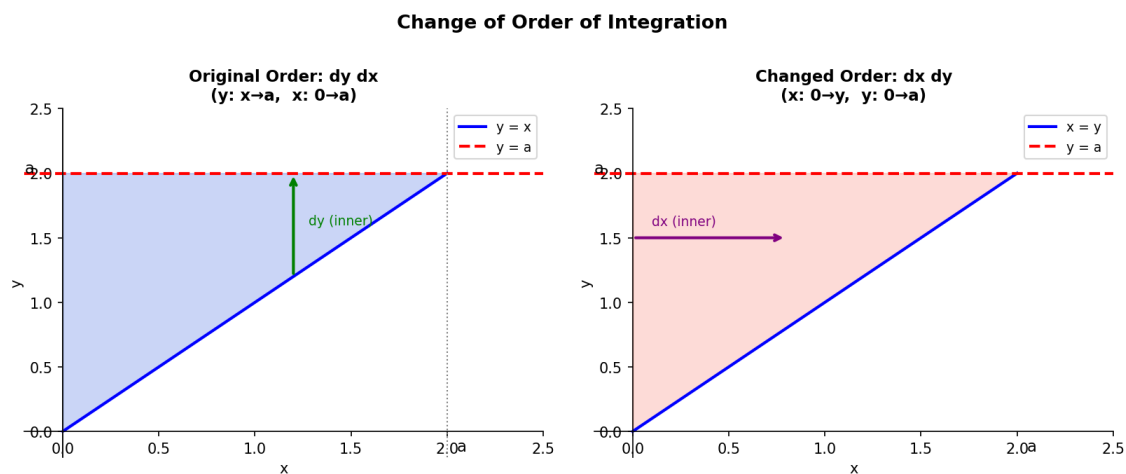


Figure 4.3 — Change of order: original $dy\,dx$ (left) changed to $dx\,dy$ (right). The shaded region R is the same — only the strip direction changes.

When the Region Splits into Two Parts

Sometimes, after changing the order, the region R cannot be described by a single set of limits. In such cases, **divide the region into two sub-regions R_1 and R_2** , evaluate the integral over each sub-region separately, and add the results.

$$\iint_R f\,dA = \iint_{R_1} f\,dA + \iint_{R_2} f\,dA$$

Solved Examples

⇒ **Example:** Change the order of integration in $\int_0^a \int_x^a f(x,y) dy dx$, then write the new integral.

Solution:

- Given limits: $y: x \rightarrow a, x: 0 \rightarrow a$
- Boundary curves: $y = x, y = a, x = 0, x = a$
- The region R is the triangle above the line $y = x$, below $y = a$, for $0 \leq x \leq a$.
- For the changed order $dx dy$: the strip is horizontal $\rightarrow x: 0 \rightarrow y, y: 0 \rightarrow a$

Changed Integral: $\int_0^a \int_0^y f(x,y) dx dy$

⇒ **Example:** Change order and evaluate $\int_0^a \int_x^a (x^2+y^2) dy dx$

Solution (using result above):

Changed integral: $\int_0^a \int_0^y (x^2+y^2) dx dy$

$$\int_0^y (x^2+y^2) dx = [x^3/3 + y^2x]_0^y = y^3/3 + y^3 = 4y^3/3$$

$$\int_0^a (4y^3/3) dy = [y^4/3]_0^a = a^4/3$$

Answer: $a^4/3$

⇒ **Example:** Change order and evaluate $\int_0^{\infty} \int_0^{\infty} (e^{-x}/y) dy dx$

Solution:

- Given: $y: x \rightarrow \infty, x: 0 \rightarrow \infty$
- Region: above the line $y = x$ in the first quadrant ($y \geq x, x \geq 0$)
- Changed order $dx dy$: $x: 0 \rightarrow y, y: 0 \rightarrow \infty$

$$\int_0^{\infty} \int_0^y (e^{-x}/y) dx dy = \int_0^{\infty} e^{-x}/y \cdot [x]_0^y dy = \int_0^{\infty} e^{-y} dy = 1$$

Answer: 1

4.5 Area Enclosed by Plane Curves (Cartesian)

Concept

The area of a region R in the xy -plane can be computed directly using a double integral. We integrate the function $f(x,y) = 1$ over the region R . Each small element of area dA contributes $1 \times dA = dA$ to the total area.

Fundamental Area Formulas

$$\text{Area} = \iint_R dA = \iint_R dy \, dx \text{ (or) } \iint_R dx \, dy$$

$$\text{Area (Cartesian)} = \int_a^b \int_{f_1(x)}^{f_2(x)} dy \, dx = \int_a^b [f_2(x) - f_1(x)] \, dx$$

How to Find Area Using Double Integration

Step 1: Find intersection points of the given curves to determine limits.

Step 2: Decide which curve is on top (or right) and which is on the bottom (or left).

Step 3: Set up the double integral with appropriate limits.

Step 4: Evaluate the inner integral first, then the outer integral.

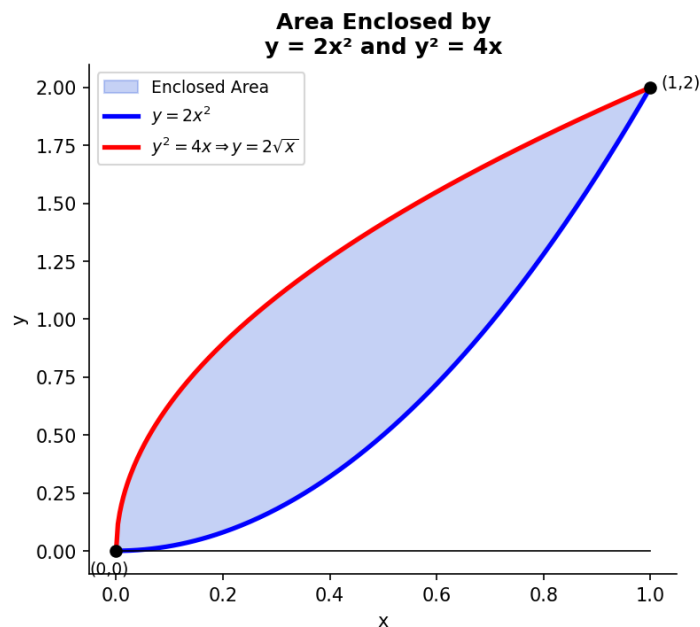


Figure 4.4 — Shaded region enclosed between $y = 2x^2$ (parabola) and $y^2 = 4x$. Area = $\int_0^1 \int_{2x^2}^{2\sqrt{x}} dy \, dx = 2/3$.

Solved Examples

— **Example:** Find the area enclosed by $y = 2x^2$ and $y^2 = 4x$.

Solution:

- Intersection: Substitute $y = 2x^2$ into $y^2 = 4x \rightarrow 4x^4 = 4x \rightarrow x = 0$ or $x = 1$
- At $x = 1$: $y = 2$. Points: $(0,0)$ and $(1,2)$
- Upper curve: $y = 2\sqrt{x}$ (from $y^2 = 4x$). Lower curve: $y = 2x^2$

$$\begin{aligned}\text{Area} &= \int_0^1 \int_{2x^2}^{2\sqrt{x}} dy \, dx = \int_0^1 (2\sqrt{x} - 2x^2) \, dx \\ &= [4x^{3/2}/3 - 2x^3/3]_0^1 = 4/3 - 2/3 = 2/3\end{aligned}$$

Answer: $2/3$ square units

→ **Example:** Find the area between parabolas $y^2 = 4ax$ and $x^2 = 4ay$.

Solution:

- Intersection: Solve simultaneously $\rightarrow x = 0$ and $x = 4a$. Points $(0,0)$ and $(4a, 4a)$.
- Upper curve: $y = 2\sqrt{ax}$. Lower curve: $y = x^2/(4a)$.

$$\begin{aligned}\text{Area} &= \int_0^{4a} \int_{x^2/(4a)}^{2\sqrt{ax}} dy \, dx = \int_0^{4a} [2\sqrt{ax} - x^2/(4a)] \, dx \\ &= [4\sqrt{a} \cdot x^{3/2}/3 - x^3/(12a)]_0^{4a} = 32a^{3/2}/3 - 16a^{3/2}/3 = 16a^{3/2}/3\end{aligned}$$

Answer: $16a^{3/2}/3$ square units

→ **Example:** Find the area of the ellipse $x^2/a^2 + y^2/b^2 = 1$.

Solution:

- By symmetry, Area = $4 \times$ (area in first quadrant)
- In first quadrant: $x: 0 \rightarrow a$, $y: 0 \rightarrow (b/a)\sqrt{a^2 - x^2}$

$$\begin{aligned}\text{Area} &= 4 \cdot (a/b) \cdot [b^2/2 \cdot \sin^{-1}(x/a) + (x/2)\sqrt{a^2 - x^2}]_0^a \\ &= 4 \cdot (b/a) \cdot (b/2 \cdot \pi/2) \cdot a/b = \pi ab\end{aligned}$$

Answer: πab square units

4.6 Area Enclosed by Plane Curves (Polar)

Polar Area Formula

When the boundary of a region is given in polar form $r = f(\theta)$, the area is calculated using the polar double integral:

$$\text{Area} = \int_{\theta_1}^{\theta_2} \int_{r_1(\theta)}^{r_2(\theta)} r \, dr \, d\theta$$

The $r \, dr \, d\theta$ element represents a small polar area element — think of it as a small "sector" of angular width $d\theta$ and radial width dr .

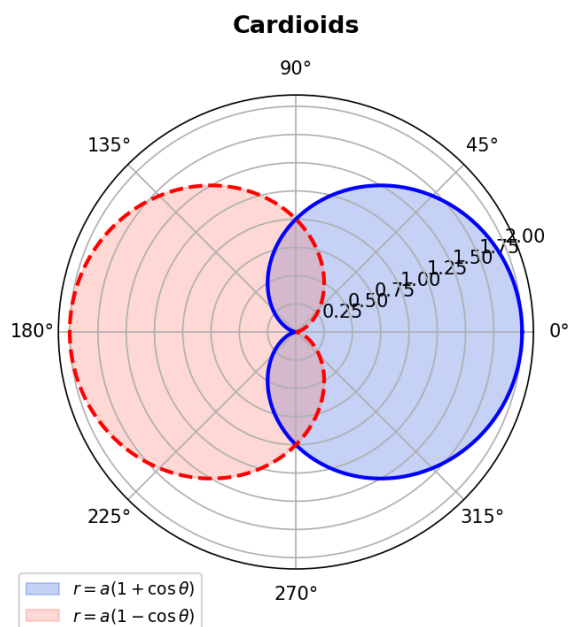


Figure 4.5 — Two cardioids: $r = a(1 + \cos\theta)$ (blue) and $r = a(1 - \cos\theta)$ (red). Both have area $= (3/2)\pi a^2$ square units.

Common Polar Curves and Their Areas

Curve	Equation	Area
Cardioid (right)	$r = a(1 + \cos\theta)$	$(3/2) \pi a^2$
Cardioid (left)	$r = a(1 - \cos\theta)$	$(3/2) \pi a^2$
Circle (polar)	$r = 2a \cos\theta$	πa^2
Lemniscate	$r^2 = a^2 \cos 2\theta$	a^2

Solved Examples

— **Example:** Find the area of the cardioid $r = a(1 + \cos\theta)$.

Solution:

- The curve is symmetric about the initial line ($\theta = 0$).
- Use symmetry: Area = $2 \times$ (area for $0 \leq \theta \leq \pi$)
- r varies from 0 to $a(1 + \cos\theta)$, θ varies from 0 to π

$$\begin{aligned}\text{Area} &= 2 \int_0^\pi \int_0^{a(1+\cos\theta)} r \, dr \, d\theta = \int_0^\pi a^2(1+\cos\theta)^2 \, d\theta \\ &= a^2 \int_0^\pi [1 + \cos^2\theta + 2\cos\theta] \, d\theta = (a^2/2) \int_0^\pi [3 + \cos 2\theta + 4\cos\theta] \, d\theta \\ &= (a^2/2)[3\theta + \sin 2\theta/2 + 4\sin\theta]_0^\pi = (a^2/2)(3\pi) = 3\pi a^2/2\end{aligned}$$

Answer: $3\pi a^2/2$ square units

— **Example:** Find the area of the lemniscate $r^2 = a^2 \cos 2\theta$.

Solution:

- The lemniscate has two loops, each symmetric about the x-axis.
- Area = $4 \times$ (area of upper half of one loop)
- For one loop: $\theta: 0 \rightarrow \pi/4$, $r: 0 \rightarrow a\sqrt{\cos 2\theta}$

$$\begin{aligned}\text{Area} &= 4 \int_0^{\pi/4} \int_0^{a\sqrt{\cos 2\theta}} r \, dr \, d\theta = 2 \int_0^{\pi/4} a^2 \cos 2\theta \, d\theta \\ &= 2a^2 [\sin 2\theta/2]_0^{\pi/4} = 2a^2 \cdot (1/2) = a^2\end{aligned}$$

Answer: a^2 square units

4.7 Triple Integrals

Definition

A **Triple Integral** is an extension of the double integral to three dimensions. It is used to integrate a function $f(x, y, z)$ over a three-dimensional region V . Physically, it computes **volume**, **mass**, **moment of inertia**, and other quantities of a 3D solid body.

$$\int_V f(x, y, z) dV = \iiint f(x, y, z) dz dy dx$$

Triple Integral – First Octant of Sphere $x^2+y^2+z^2=a^2$

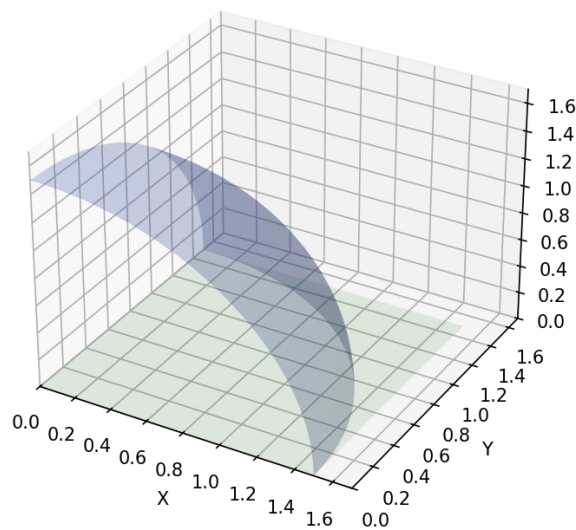


Figure 4.6 — First octant of sphere $x^2+y^2+z^2=a^2$. The triple integral over this region gives its volume as $(1/8) \cdot (4\pi a^3/3)$.

How to Evaluate a Triple Integral

Step 1: Determine the limits for z (innermost), then y , then x (outermost).

Step 2: Integrate with respect to z first (treating x and y as constants). Substitute z -limits.

Step 3: Integrate the result with respect to y (treating x as constant). Substitute y -limits.

Step 4: Integrate the result with respect to x . Substitute x -limits to get the final answer.

Solved Examples

→ **Example:** Evaluate $\int_0^1 \int_0^1 \int_0^1 (x+y+z) dz dy dx$

Solution:

$$\text{Inner (dz): } \int_0^1 (x+y+z) dz = [xz+yz+z^2/2]_0^1 = xc+yc+c^2/2$$

$$\text{Middle (dy): } \int_0^1 (xc+yc+c^2/2) dy = [xcy+cy^2/2+c^2y/2]_0^1 = xbc+b^2c/2+bc^2/2$$

$$\text{Outer (dx): } \int_0^1 (xbc+b^2c/2+bc^2/2) dx = [a^2bc/2+ab^2c/2+abc^2/2]$$

$$= abc(a+b+c)/2$$

Answer: $abc(a+b+c)/2$

⇒ **Example:** Find $\iiint xyz \, dz \, dy \, dx$ over the positive octant of $x^2+y^2+z^2=a^2$.

Solution:

- Limits: $z: 0 \rightarrow \sqrt{a^2-x^2-y^2}$, $y: 0 \rightarrow \sqrt{a^2-x^2}$, $x: 0 \rightarrow a$

After integrating w.r.t. z then y then x: Answer = $a^4/48$

Answer: $a^4/48$

4.8 Volume of Solids Using Triple Integrals

Concept

Just as area is computed by integrating 1 over a 2D region, **volume** is computed by integrating 1 over a 3D region. The triple integral of $f(x,y,z) = 1$ over a volume V gives the volume of that solid body.

$$\text{Volume} = \iiint_V dV = \iiint_V dz \, dy \, dx$$

Standard Volume Results

Solid	Volume Formula	Result
Sphere $x^2+y^2+z^2=a^2$	8 × first-octant integral	$4\pi a^3/3$
Ellipsoid $x^2/a^2+y^2/b^2+z^2/c^2=1$	Similar to sphere	$4\pi abc/3$
Tetrahedron ($x/a+y/b+z/c=1$)	$\iiint dz \, dy \, dx$	$abc/6$

Solved Examples

⇒ **Example:** Find the volume of the sphere $x^2+y^2+z^2=a^2$.

Solution:

- By symmetry: Volume = 8 × (first octant volume)
- Limits in first octant: $z: 0 \rightarrow \sqrt{a^2-x^2-y^2}$, $y: 0 \rightarrow \sqrt{a^2-x^2}$, $x: 0 \rightarrow a$

$$V = 8 \int_0^a \int_0^{\sqrt{a^2-x^2}} \int_0^{\sqrt{a^2-x^2-y^2}} dz \, dy \, dx$$

$$\text{After integration: } V = 4\pi a^3/3$$

Answer: $V = 4\pi a^3/3$ cubic units

⇒ **Example:** Find the volume of the tetrahedron bounded by $x/a + y/b + z/c = 1$ and coordinate planes.

Solution:

- Limits: $z: 0 \rightarrow c(1 - x/a - y/b)$, $y: 0 \rightarrow b(1 - x/a)$, $x: 0 \rightarrow a$

$$V = \int_0^a \int_0^{b(1-x/a)} \int_0^{c(1-x/a-y/b)} dz \, dy \, dx$$

$$\text{After double integration: } V = abc/6$$

Answer: $V = abc/6$ cubic units

4.9 Change of Variables in Integrals

4.9(a) Cartesian to Polar Co-ordinates (Double Integral)

When the region of integration is circular or contains terms like x^2+y^2 , converting to polar co-ordinates simplifies the integral significantly.

Transformation Substitutions:

$$x = r \cos\theta, y = r \sin\theta, x^2 + y^2 = r^2$$

$$dx dy \rightarrow r dr d\theta \text{ (Jacobian = } r \text{)}$$

Working Rule:

- Step 1:** Check the order of integration (make it correct if needed).
- Step 2:** Write the boundary equations from the limits.
- Step 3:** Sketch the region of integration.
- Step 4:** Apply the substitution: $x = r \cos\theta, y = r \sin\theta, x^2+y^2 = r^2, dx dy = r dr d\theta$.
- Step 5:** Determine new limits for r and θ from the sketch (draw radial strip inside the region).
- Step 6:** Evaluate the transformed polar integral.

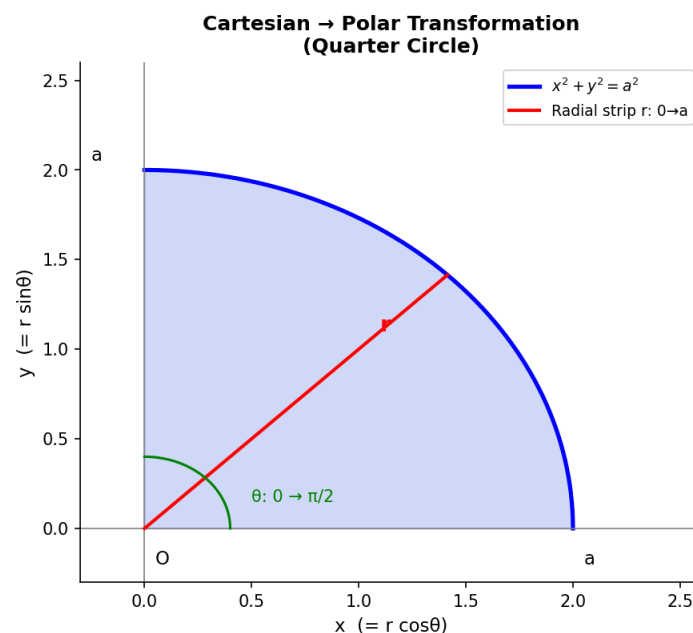


Figure 4.7 — Converting the quarter-circle region (Cartesian) to polar form. r varies from 0 to a , θ varies from 0 to $\pi/2$.

Solved Examples – Cartesian to Polar

— **Example:** Evaluate $\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} e^{-(x^2+y^2)} dy dx$ using polar co-ordinates.

Solution:

- Region: first quadrant ($x \geq 0, y \geq 0$). In polar: $\theta: 0 \rightarrow \pi/2, r: 0 \rightarrow \infty$

$$\int_0^{\pi/2} \int_0^{\infty} e^{-r^2} \cdot r \, dr \, d\theta$$

- Let $t = r^2$, $dt = 2r \, dr \rightarrow r \, dr = dt/2$

$$= \int_0^{\pi/2} d\theta \cdot \int_0^{\infty} e^{-t} \frac{dt}{2} = \left(\frac{\pi}{2}\right) \cdot \left(\frac{1}{2}\right) = \pi/4$$

Answer: $\pi/4$

■ **Note:** This result is used to prove that $\int_0^{\infty} e^{-x^2} dx = \sqrt{\pi}/2$ — a fundamental result in probability!

⇒ **Example:** Evaluate $\int_0^{2a} \int_0^{\sqrt{2ax-x^2}} (x^2+y^2) \, dy \, dx$ by converting to polar.

Solution:

- Boundary: $y = \sqrt{2ax - x^2} \rightarrow y^2 = 2ax - x^2 \rightarrow x^2 + y^2 - 2ax = 0 \rightarrow$ circle with centre $(a,0)$ radius a .
- In polar: $r = 2a \cos\theta$. Limits: $r: 0 \rightarrow 2a \cos\theta$, $\theta: 0 \rightarrow \pi/2$

$$\begin{aligned} \int_0^{\pi/2} \int_0^{2a \cos\theta} r^2 \cdot r \, dr \, d\theta &= \int_0^{\pi/2} \int_0^{2a \cos\theta} r^3 \, dr \, d\theta \\ &= \int_0^{\pi/2} \left[\frac{r^4}{4} \right]_0^{2a \cos\theta} d\theta = 4a^4 \int_0^{\pi/2} \cos^4\theta \, d\theta \\ &= 4a^4 \cdot \left(\frac{3}{4}\right) \cdot \left(\frac{1}{2}\right) \cdot \left(\frac{\pi}{2}\right) = 3\pi a^4/4 \end{aligned}$$

Answer: $3\pi a^4/4$

4.9(b) Cylindrical Co-ordinates

Cylindrical co-ordinates are used when the region has **circular cross-sections at every height z** — like cylinders, cones, and paraboloids. It is essentially polar co-ordinates in the x - y plane with z unchanged.

Transformation:

$$x = r \cos\theta, y = r \sin\theta, z = z$$

$$dx \, dy \, dz \rightarrow r \, dr \, d\theta \, dz \text{ (Jacobian = } r \text{)}$$

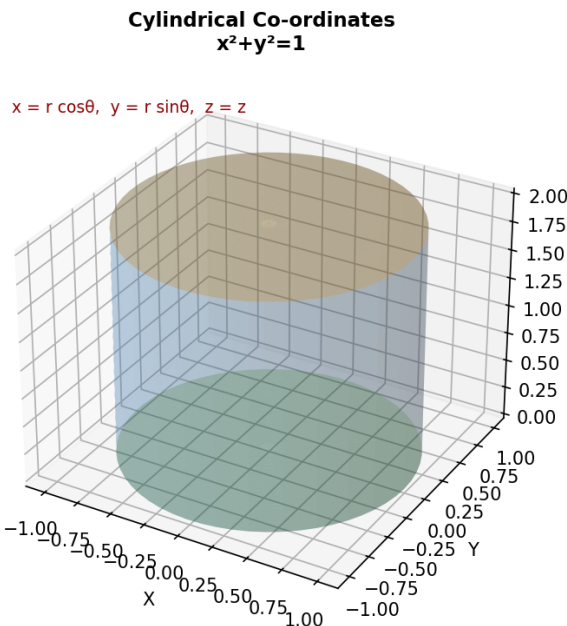


Figure 4.8 — Cylinder $x^2 + y^2 = 1$, $z = 0$ to 2 in cylindrical co-ordinates. The green base and orange top cap are clearly visible.

When to use Cylindrical Co-ordinates?

- The region involves a **cylinder** ($x^2 + y^2 = a^2$) or a **paraboloid** ($z = x^2 + y^2$).
- The integrand contains the expression $x^2 + y^2$.
- The cross-section parallel to the xy -plane is **circular**.

Solved Example – Cylindrical Co-ordinates

— **Example:** Find the volume of the cylinder $x^2 + y^2 = 1$ between $z = 0$ and $z = 4 - r^2$.

Solution:

- Convert: $r^2 = 1$, so $r: 0 \rightarrow 1$; $z: 0 \rightarrow 4 - r^2$; $\theta: 0 \rightarrow 2\pi$

$$\begin{aligned} V &= \int_0^{2\pi} \int_0^1 \int_0^{4-r^2} r \, dz \, dr \, d\theta \\ &= \int_0^{2\pi} \int_0^1 r(4-r^2) \, dr \, d\theta = \int_0^{2\pi} [2r^2 - r^3/4]_0^1 \, d\theta \\ &= \int_0^{2\pi} (7/4) \, d\theta = 7\pi/2 \end{aligned}$$

Answer: $7\pi/2$ cubic units

4.9(c) Spherical Polar Co-ordinates

Spherical co-ordinates are ideal when the region is **spherical or has spherical symmetry**. The point (x,y,z) is described by its distance r from the origin, polar angle θ from the z -axis, and azimuthal angle ϕ in the xy -plane.

Transformation:

$$x = r \sin\theta \cos\phi, y = r \sin\theta \sin\phi, z = r \cos\theta$$

$$dx dy dz \rightarrow r^2 \sin\theta dr d\theta d\phi \text{ (Jacobian} = r^2 \sin\theta)$$

Spherical Polar Co-ordinates
 $x=r \sin\theta \cos\phi, y=r \sin\theta \sin\phi, z=r \cos\theta$

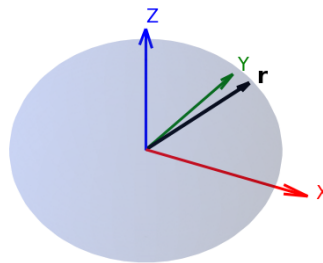


Figure 4.9 — Spherical polar co-ordinates: r = distance from origin, θ = polar angle from z -axis, ϕ = azimuthal angle in xy -plane.

Standard Limits (Full Sphere):

$$r: 0 \rightarrow a, \theta: 0 \rightarrow \pi, \phi: 0 \rightarrow 2\pi$$

Solved Example – Spherical Co-ordinates

→ **Example:** Evaluate $\int \frac{1}{\sqrt{1-x^2-y^2-z^2}} dV$ over the sphere $x^2+y^2+z^2=1$.

Solution:

- Convert: $1-x^2-y^2-z^2 = 1-r^2$, $dV = r^2 \sin\theta dr d\theta d\phi$
- Limits: $r: 0 \rightarrow 1, \theta: 0 \rightarrow \pi, \phi: 0 \rightarrow 2\pi$

$$\begin{aligned} &= \int_0^{2\pi} d\phi \cdot \int_0^\pi \sin\theta d\theta \cdot \int_0^1 \frac{r^2}{\sqrt{1-r^2}} dr \\ &= 2\pi \times 2 \times \int_0^1 \frac{r^2}{\sqrt{1-r^2}} dr \end{aligned}$$

- Put $r = \sin t$, $dr = \cos t dt \rightarrow \int_0^{\pi/2} \sin^2 t dt = \pi/4$

$$= 4\pi \times (\pi/4) = \pi^2$$

Answer: π^2

Advantages and Disadvantages of Key Methods

Double Integration – Cartesian Co-ordinates

Advantages:

- Simple and straightforward to apply when limits are constants or simple algebraic functions.
- Easy to visualize regions as rectangular or triangular shapes in the xy -plane.
- Direct application to problems involving rectangular domains without any substitution.
- Both orders $dy\,dx$ and $dx\,dy$ can be chosen based on which gives simpler limits.

Disadvantages:

- Becomes very complicated when the region is circular or has curved boundaries — limits involve square roots.
- The integrand may contain x^2+y^2 terms that do not simplify easily in Cartesian form.
- For problems involving circles or ellipses, the limits become functions, making evaluation tedious.
- Not suitable for problems that are symmetric about a point (radial symmetry), as polar form is more natural.

Double Integration – Polar Co-ordinates

Advantages:

- Extremely convenient for circular, annular, or radially symmetric regions — limits become simple constants.
- Integrand containing x^2+y^2 simplifies immediately to r^2 .
- Area of cardioids, lemniscates, and other polar curves is easily computed.
- Used in evaluating important integrals like $\int_{-\infty}^{\infty} e^{-x^2} dx = \sqrt{\pi}/2$.

Disadvantages:

- Requires familiarity with polar curve sketching — a common source of errors.
- The Jacobian factor " r " must always be included; forgetting it is a frequent mistake.
- Limits can become non-trivial when the region is not purely radially symmetric.
- Not suitable for regions with straight-line boundaries (triangles, squares), where Cartesian is simpler.

Change of Order of Integration

Advantages:

- Converts a difficult or impossible integral into an easily evaluable one.
- Allows us to evaluate integrals where the inner integral has no closed form (e.g., $\sin(x)/x$).
- Helps evaluate double integrals over complex regions by splitting into simpler sub-regions.

- Important technique for MSBTE examinations — saves time when the original order is difficult.

Disadvantages:

- Requires careful region sketching — an incorrect sketch leads to wrong limits.
- When the region splits into R_1 and R_2 , both sub-regions must be handled separately, increasing work.
- Not always obvious whether changing the order will simplify the problem.

Triple Integrals / Change of Variables

Advantages:

- Triple integrals directly give volume, mass, moment, and centre of gravity of 3D solids.
- Cylindrical co-ordinates make cylinder and cone problems extremely simple to set up.
- Spherical co-ordinates reduce sphere problems to integrating simple trigonometric functions.
- The Jacobian transformation is a systematic and reliable way to change variables in any dimension.

Disadvantages:

- The Jacobian (r for cylindrical, $r^2 \sin\theta$ for spherical) must be included — easy to forget.
- Visualizing 3D regions requires spatial thinking, which is more challenging than 2D regions.
- Setting up correct limits in 3D (especially for odd shapes) requires careful analysis.
- Triple integrals with variable limits involve much more algebraic computation than double integrals.

Quick Reference Formula Sheet

All Key Formulas at a Glance

Topic	Formula / Key Result
Double Integral (Cartesian)	$\int_a^b \int_{f(x)}^{g(x)} f(x,y) dy dx$
Separable Integral	$\int_a^b \int_c^d f(x)g(y) dy dx = [\int_a^b f(x) dx][\int_c^d g(y) dy]$
Polar Double Integral	$\iint f(r,\theta) r dr d\theta$ (note: Jacobian = r)
Cartesian $\rightarrow$ Polar	$x=r \cos\theta, y=r \sin\theta, dx dy = r dr d\theta$
Area (Cartesian)	$\iint_R dA = \iint dy dx$
Area (Polar)	$\iint r dr d\theta$
Area of Ellipse	πab
Area of Cardioid $r=a(1\pm\cos\theta)$	$3\pi a^2/2$
Area of Circle $r=2a \cos\theta$	πa^2
Area of Lemniscate $r^2=a^2\cos 2\theta$	a^2
Triple Integral	$\iiint_V f dV = \iiint f dz dy dx$
Volume of Sphere	$4\pi a^3/3$
Volume of Ellipsoid	$4\pi abc/3$
Volume of Tetrahedron	$abc/6$
Cylindrical Jacobian	$J = r$ ($dx dy dz = r dr d\theta dz$)
Cylindrical Transform	$x=r \cos\theta, y=r \sin\theta, z=z$
Spherical Jacobian	$J = r^2 \sin\theta$ ($dx dy dz = r^2 \sin\theta dr d\theta d\phi$)
Spherical Transform	$x=r \sin\theta \cos\phi, y=r \sin\theta \sin\phi, z=r \cos\theta$
Walli's: $\int_0^{\pi/2} \sin^n \theta$ or $\cos^n \theta$ (n even)	$[(n-1)/n \cdot (n-3)/(n-2) \cdot \dots \cdot 1/2] \cdot \pi/2$
Walli's: $\int_0^{\pi/2} \sin^n \theta$ or $\cos^n \theta$ (n odd)	$[(n-1)/n \cdot (n-3)/(n-2) \cdot \dots \cdot 2/3]$
Famous Result	$\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} e^{-(x^2+y^2)} dy dx = \pi/4$

Important Notes for MSBTE Examinations

1. Always **check the correct form** of the integral before evaluating — inner limits must match the inner variable.
2. For **polar integrals**, never forget the extra factor "**r**" (Jacobian). This is the most common error.
3. For **Change of Order**, always draw the region diagram first. The limits follow directly from the diagram.

4. When using **symmetry** (e.g., for cardioids), multiply the half-region result by 2 or 4 as appropriate.
5. The **Walli's formula** is essential for evaluating trigonometric integrals in polar and spherical problems.
6. For **triple integrals**, set up limits from innermost (z) to outermost (x) — z first, y second, x last.
7. Volume of sphere = $\frac{4\pi a^3}{3}$, volume of ellipsoid = $\frac{4\pi abc}{3}$ — memorise these!
8. The **Jacobian** for cylindrical = r ; for spherical = $r^2 \sin\theta$. These must be included as extra factors.

Best of Luck in Your MSBTE K-Scheme Examinations! ■